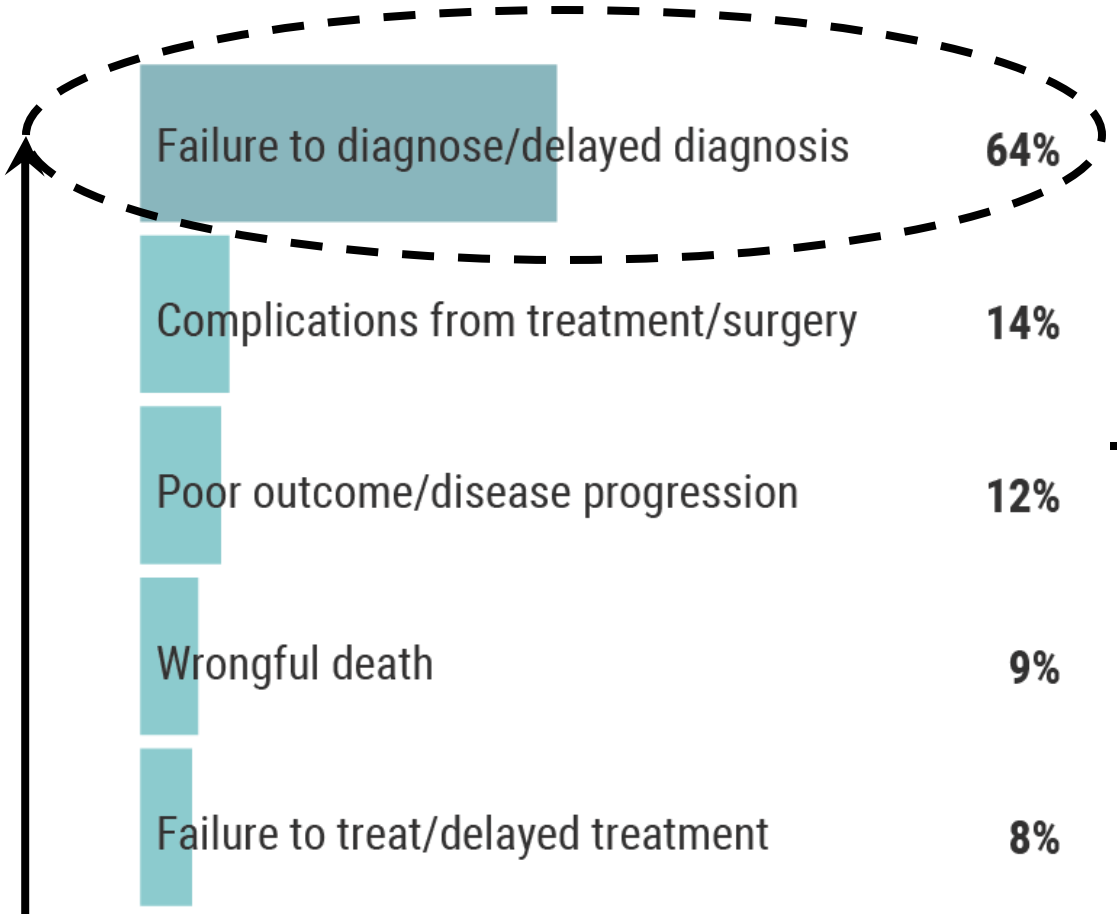
Embraine

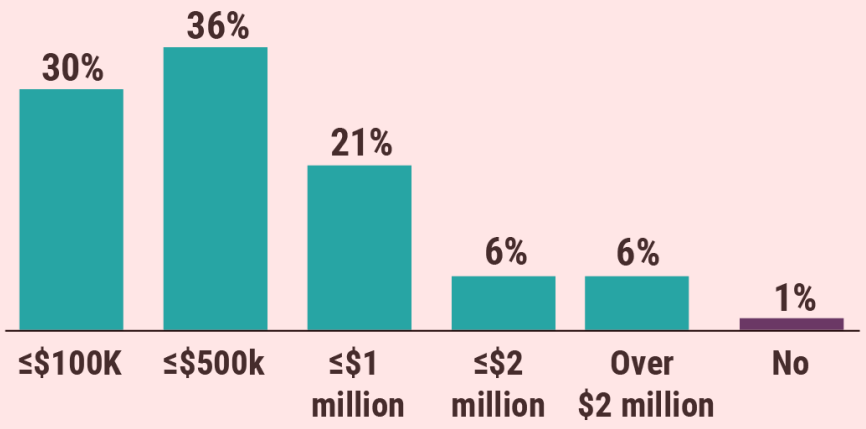
A NeuroVNN-based AI software for aiding brain MRI-based health assessments

Radiologists face high risk of litigation[a]



Financial Impacts

- Most lawsuits result in a **payout** of $100k or more



- **Malpractice insurance coverage** costs more after lawsuit (~30% pay $20,000 or more annually)

Reasons

- Radiologists are typically **over-worked**
- Performance depends on **human capabilities** (visual detection, cognitive reasoning, experience)

[a] https://www.medscape.com/slideshow/2021-malpractice-report-radiologist-6014797

AI has shown much promise in research, but challenges for practical adoption exist…

Past two decades of AI research has shown that

- AI can bring **automation** to radiology workflows
- AI methods are **more sensitive** to detecting pathology patterns than radiologists
- AI provides a **data-centric assessment** of MRI

Black-box

Generalizability to patient populations

Challenges to practical adoption of AI

Reproducibility

NEED

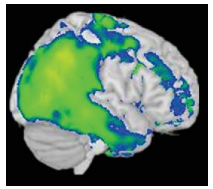
credible AI-based algorithms (robust, transparent, generalizable)

for assessing brain MRI to reduce the error rate of radiologists

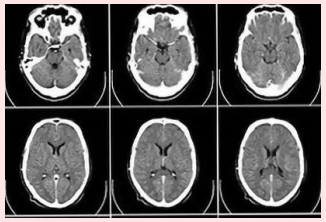
NeuroVNN: Proposed solution based on coVariance Neural Networks (VNNs)

- VNNs provide robust, explainable, generalizable AI solutions [b]
- NeuroVNN is **pre-trained** on brain MRI data from healthy population
- NeuroVNN can identify **atrophy-based biomarkers** in brain MRI for a variety of health conditions: Foundation model (ChatGPT) for biomarkers in MRI
- Different health conditions may have **common clinical symptoms but different patterns of atrophy** [c] (or even no pathological atrophy)

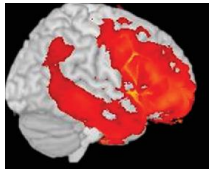
Alzheimer's Disease



clinical symptoms (decreased motivation, trouble concentrating, etc.) + Brain imaging



Frontotemporal Dementia



Clinical Depression

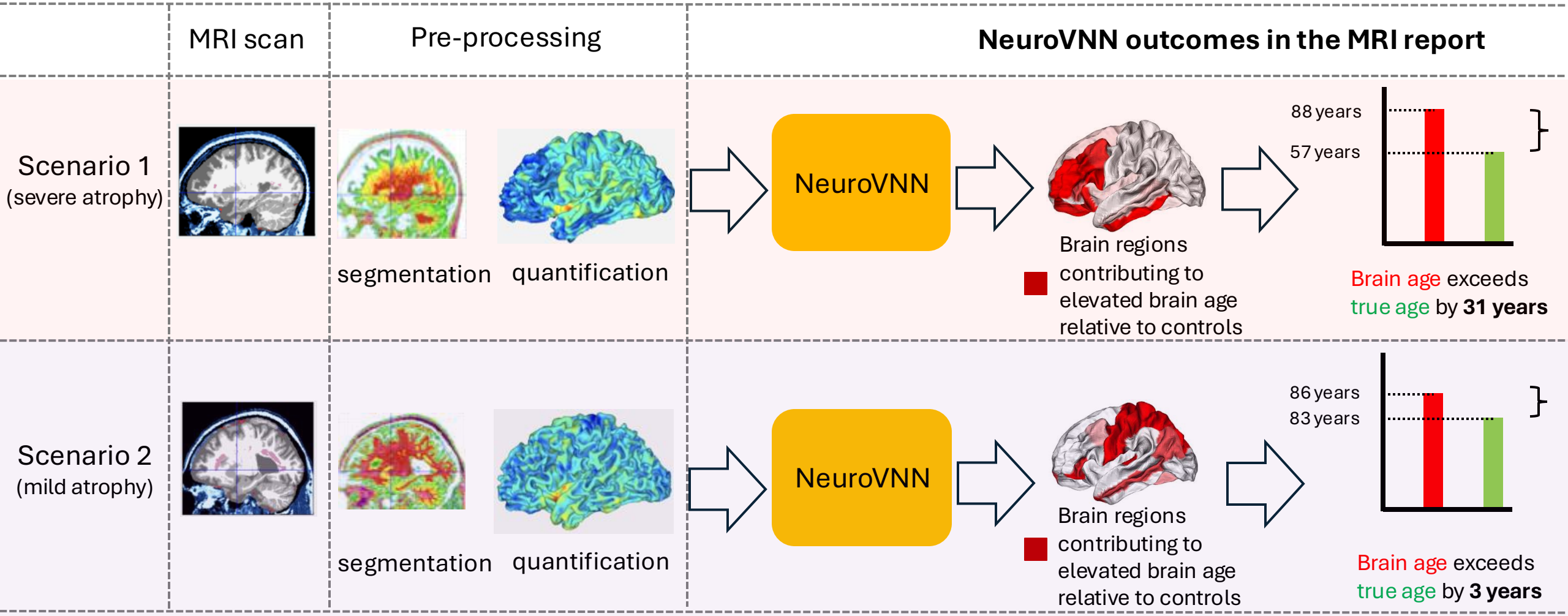
[b] Sihag, Saurabh, et al. "coVariance Neural Networks." *NeurIPS 2022*

[c] Rabinovici, Gil D. et al. "Distinct MRI Atrophy Patterns in Autopsy-Proven Alzheimer's Disease and Frontotemporal Lobar Degeneration." *American Journal of Alzheimer's Disease and Other Dementias* (2008).

NeuroVNN provides brain health assessment based on abnormal atrophy patterns in brain MRI

Product Example: 'Brain age' from MRI quantifies biological age [d]; brain age is sensitive to atrophy

Brain age predicted by NeuroVNN ∝ individual risks or severity of adverse health conditions

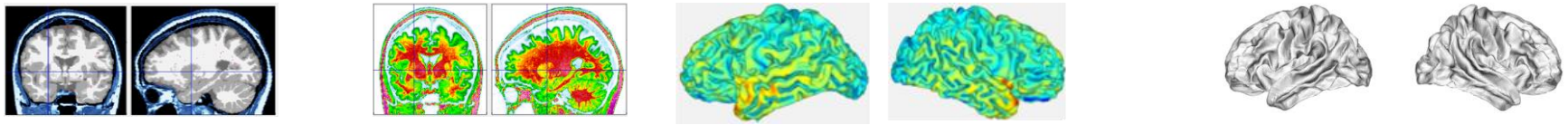


[d] Sihag, Saurabh, et al. "Explainable Brain Age Prediction using coVariance Neural Networks." *NeurIPS 2023*

NeuroVNN provides brain health assessment based on abnormal atrophy patterns in brain MRI

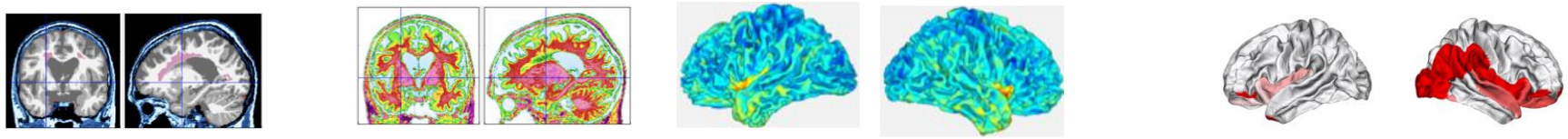
Example 1: A cognitively **healthy** individual (age: 55 years)

NeuroVNN outputs



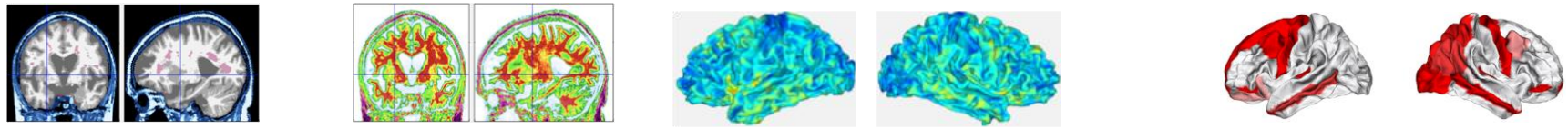
Brain age = 55.1 years

Example 2: An individual diagnosed with **progressive non-fluent aphasia** (age: 77 years)



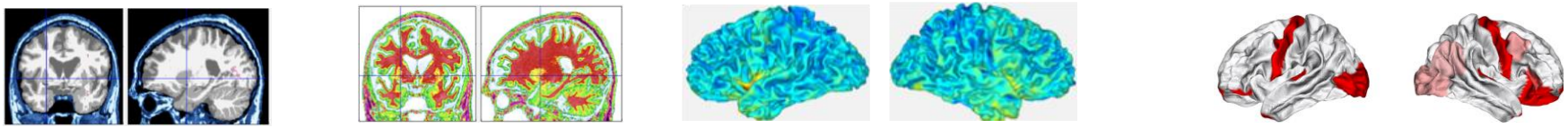
Brain age = 84 years

Example 3: An individual diagnosed with **Alzheimer's disease** (age: 84 years)



Brain age = 88 years

Example 4: An individual diagnosed with **mild cognitive impairment** (age: 75 years)

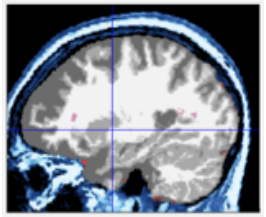


Brain age = 87 years

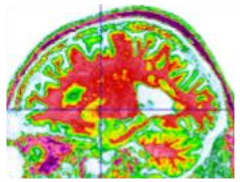
Brain regions contributing to elevated brain age

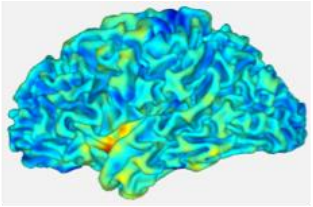
NeuroVNN adds data-driven sensitivity and specificity to MRI assessment for a radiologist

- Most radiologists rely on visual inspection for MRI assessment; market competitors offer quantification of MRI



MRI scan





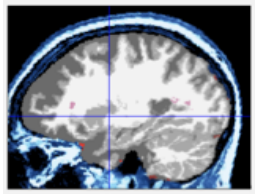
NeuroQuant® Neuroreader®

VUNO

MRI quantification adds redundancy of information

Adds limited diagnostic value (still prone to human errors)

- Typical deep learning approaches provide (opaque) assessment of brain health from MRI



MRI scan

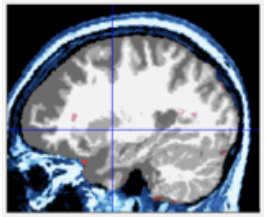
AI-based decision on the health of the brain

Decision cannot be explained

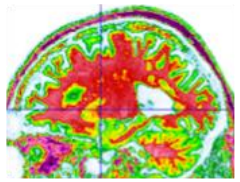
Adds automation but no diagnostic value

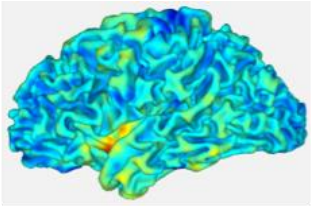
NeuroVNN adds data-driven sensitivity and specificity to MRI assessment for a radiologist

- Most radiologists rely on visual inspection for MRI assessment; market competitors offer quantification of MRI



MRI scan





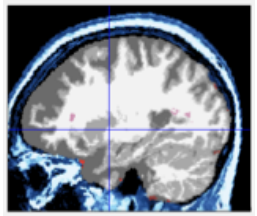
NeuroQuant® Neuroreader®

VUNO

MRI quantification adds redundancy of information

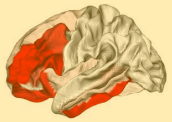
Adds limited diagnostic value (still prone to human errors)

- NeuroVNN provides **transparent assessment** of brain health from MRI



MRI scan

NeuroVNN-based decision on the health of the brain

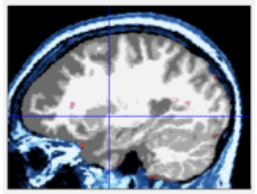


Explanation of decision

NeuroVNN provides verifiable biological sensitivity for any health condition with atrophy

Adds automation and diagnostic value

- Typical deep learning approaches provide (opaque) assessment of brain health from MRI



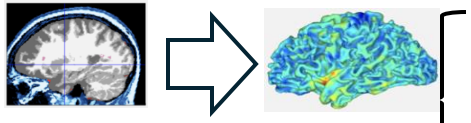
MRI scan

AI-based decision on the health of the brain

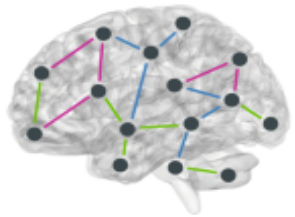
Decision cannot be explained

Adds automation but no diagnostic value

Competition



MRI quantification

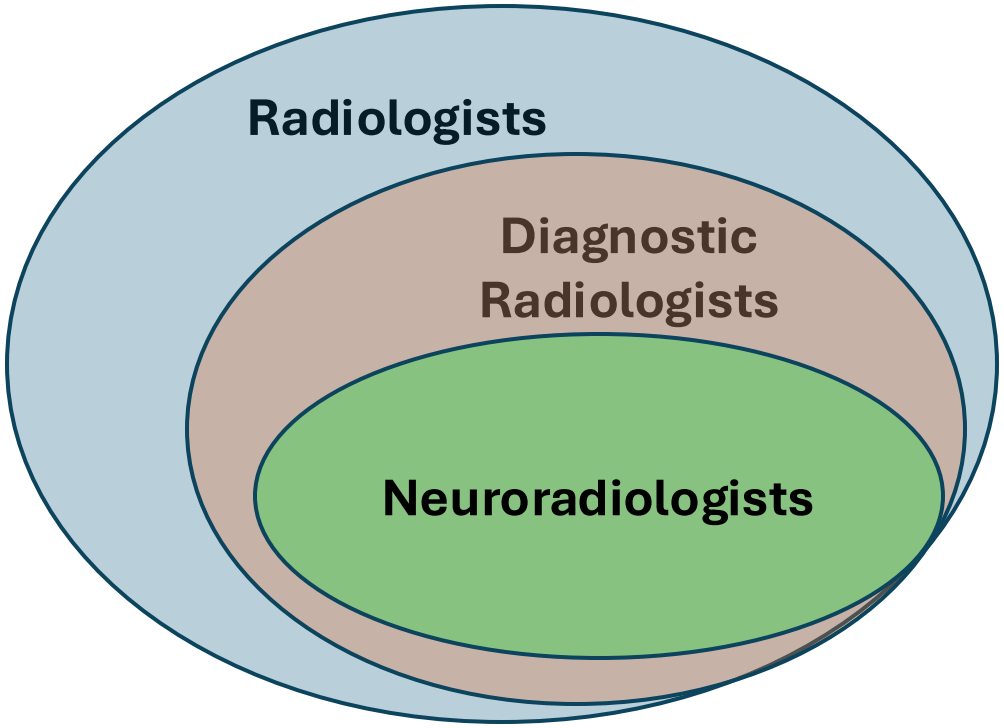


Leverage graph-theoretic, biological patterns in MRI data

Solution Category	Solution Option	Biologically Plausible	Reproducible	Transparent	Informative
Commercial Competitors (non-AI)	NeuroQuant® VUNO Neuroreader®	++	++	++	- -
Traditional data-centric approaches	Principal Component Analysis	+	- -	++	+
	Support Vector Regression	+	- -	++	+
Deep learning approaches	FC neural networks	-	-	--	+
	Convolutional neural networks	-	+	-	++
VNN approach	NeuroVNN	++	++	++	++

--	-	+	++
Feature missing	Feature missing; research ongoing	Feature exists with uncertainty	Feature exists with high confidence

Market size: Two viable business models



Market	Subscription model* ($10,000/yr)		Pay-per-MRI scan model† ($100/MRI scan)	
	USA** (n.r.)	USA ($/year)	USA†† (n.s.)	USA ($/year)
Radiologists	50,137	$501M	~8M	$800M
Diagnostic Radiologists	27,197	$272M	~4.34M	$434M
Neuro-radiologists	4,311	$43M	~2.5M	$250M

n.r.: number of individuals
n.s.: number of brain MRI scans

Subscription model

* estimate based on dementia-related cases reviewed by neuroradiologists at Penn within last year (03/2023-03/2024)

** Data from Association of American Medical Colleges and Statista

Pay-per-MRI scan model

Note: Effective Jan. 2024, CPT codes 0865T and 0866T that support the use of software to analyze brain MRI have been introduced, providing pathways for reimbursement from insurance companies

† Competitor NeuroQuant adds a cost of $80-200/MRI scan for assessment

†† 40M total MRI scans/year in USA (source: Statista), we estimate about 20% of them to be brain MRI scans for radiologists and diagnostic radiologists

Path to market

Spring 2025	Fall 2025	2026	2027
Proof of Concept	Product Design	Product Development	Product Launch
Algorithm development ✓ Validation on multiple clinical datasets ✓ File IP ✓ Form an LLC ✓	User interface Integration with MRI machines Software design for fast processing FDA pre-submis-sion	FDA submission Get inputs from radiologists and neurologists Consultants for data and patient privacy	FDA approval Sales and marketing Data servers Partnerships with hospitals and imaging centers

Cross-disciplinary team

Alejandro Ribeiro,
Professor, ESE,
UPenn

Colbey Freeman
Radiologist,
Penn School of Medicine

Saurabh Sihag
Assistant Professor, ECE
SUNY Albany

NeuroVNN, integrated within MRI PACS for radiologists, will lead to more accurate diagnostic decisions